Practical No 03

A) Create methods add(), multiply(), subtract(), divide, with suitable parameters and call these methods using concept of C# delegate.

Project Name: Arithmetic

Project Type: Console app

Program Code:

I.Filename: Program.cs

```
using System;

namespace Arithmatic
{
    public delegate void Calculation(int a, int b);
    class Arithmetic
    {
        public void Add(int a,int b)
        {
            Console.WriteLine("Addition=" + (a + b));
        }
        public void Subtract(int a,int b)
        {
            Console.WriteLine("Subtraction=" + (a - b));
        }
        public void Multiply(int a, int b)
        {
            Console.WriteLine("Multiplication=" + (a * b));
        }
        public void Divide(int a, int b)
        {
            Console.WriteLine("Division=" + (a / b));
        }
    }
    class Program
    {
        static void Main()
        {
            Arithmetic obj = new Arithmetic();
            Calculation c;
            c = obj.Add;
            c(20, 10);
            c = obj.Subtract;
            c(20, 10);
            c = obj.Multiply;
            c(20, 10);
            c = obj.Divide;
            c(20, 10);
            Console.ReadLine();
        }
    }
}
```

B) Write a program using multicast delegate

Project Name: Multicast

Project Type: Console App

Program Code:

I.Filename: Program.cs

```
using System;

public delegate void Demo();

class Test
{
    public void Show()
    {
        Console.WriteLine("show method called");
    }

    public void Display()
    {
        Console.WriteLine("display method called");
    }

    public void Print()
    {
        Console.WriteLine("print method called");
    }
}

class Program
{
    static void Main()
    {
        Test t = new Test();
        Demo d;

        d = t.Show;
        d += t.Display;
        d += t.Print;

        d();

        Console.ReadLine();
    }
}
```

C) Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Project Name: delc

Project Type: Console App

I.Filename: Program.cs

```
using System;
namespace delc
{
    class BankAccount
    {
        private double balance;

        public double Balance
        {
            get
            {
                return balance;
            }
            set
            {
                if (value >= 0)
                {
                    balance = value;
                }
                else
                {
                    Console.WriteLine("Balance cannot be negative");
                }
            }
        }
        public int AccountNumber { get; set; }
    }
    class Program
    {
        static void Main(string[] args)
        {
            BankAccount b = new BankAccount();
            b.AccountNumber = 101;
            b.Balance = 5000;
            Console.WriteLine("Account Number: " + b.AccountNumber);
            Console.WriteLine("Balance: " + b.Balance);
            b.Balance = -2000;

            Console.ReadLine();
        }
    }
}
```